

Mehedi Hasan

✉ mmehedihasan29088@gmail.com 🌐 My Portfolio 🔄 mehedihasan88 🔗 mehedi-hasan88 📍 Dhaka, Bangladesh

EDUCATION

Jahangirnagar University
MSc. in Computer Science & Engineering

Aug 2023 – Sep 2024
GPA: 3.50/4.0

Jahangirnagar University
BSc. in Computer Science & Engineering

Feb 2018 – July 2023
GPA: 3.47/4.0

EXPERIENCE

Samsung R&D Institute Bangladesh
Senior Software Engineer

Dhaka, Bangladesh
Jan 2024 – Present

Innovation & Research | *Key Fields: ML, NLP, Agentic AI, XAI, Information Retrieval, Wearable Devices*

- **Developed an interactive prior-art search agent for patent analysis** that refines engineers' natural-language descriptions of ideas, retrieves relevant prior-art documents, and identifies potential multi-patent obviousness relationships. Designed an **adaptive knowledge graph and generator-validator framework** to connect related concepts across patents, validate analyses against supporting evidence, and explain why particular prior art is relevant to the user's idea.
- Fine-tuned an on-device model for Samsung Galaxy Watch for contextual positioning of the floating ongoing icon on the watchface, improving detection accuracy by approximately 21% over the base model.
- Proposed the initial idea and contributed to the development of a novel machine-learning methodology for liquid consumption estimation using wearable devices.
- Developed an automated weekly-report summarization pipeline from GitHub commit histories to support engineering workflow analysis.
- Fine-tuned a **Bangla Named Entity Recognition model** and deployed it on a mobile device.
- Contribute to team innovation reviews by evaluating proposed ideas and providing technical feedback on **novelty, obviousness, feasibility, and potential prior-art relationships**.

Development | *Tech Stack: Java, Kotlin, Jetpack Compose, Android, Hilt, WearOS*

- Developed core modules including notifications, Nowbar, watchface overlay, and backup-and-restore for **Samsung Galaxy Watch 7/8**, while prototyping new features and supporting junior engineers.
- Designed the backend architecture for Nowbar using concurrent data structures, integrated the AOD overlay system, and reduced settings-module latency from approximately 30ms to 5ms through an MVVM-based redesign.
- Developed a proof-of-concept link-preview system for the watch notification panel, identified potential security vulnerabilities, and proposed corresponding mitigation strategies.

PUBLICATIONS

- **Mehedi Hasan, Mahbub Islam Mahim, Jugal Krishna Das (2026). Segmentation Strategies for Long-Code Authorship Detection: Balancing Context Coverage and Computational Efficiency.** | *Accepted*
 - Studied how transformer-based authorship detection degrades when long source code exceeds the model context window, causing stylistic and structural information to be lost.
 - Compared truncation, multi-view, and sliding-window strategies for long-code representation, highlighting trade-offs between context coverage, classification performance, and computational cost.
- **Mehedi Hasan, Mahbub Islam Mahim (2025). Divide and Conquer: A Two-Stage Label-Wise Cascaded Framework for Bangla Hate Speech Detection.** | **Code**  **Paper**  | **Best Paper Award** 
 - Proposed a cascaded framework combining binary and multiclass models with k-fold cross-validation and intra-label ensembling to address low-resource Bangla hate-speech classification.
 - Achieved 1st place in both target identification and joint type-target-severity classification, with micro-F1 scores of 0.7356 and 0.7392, respectively, in the BLP Workshop 2025 Shared Task.

SELECTED PROJECTS

- **Human–Machine–Hybrid–Adversarial Code Classification** [↗](#) – *PyTorch, scikit-learn, NumPy, Pandas*
 - Developed an adaptive training loop that identifies difficult classes and repeatedly misclassified code samples after each training round, then increases their representation in subsequent rounds to improve performance under class imbalance.
 - Explored AST-based features, overlapping-window analysis, and multi-view learning to capture program-level information beyond local token patterns.
- **Layerwise Probing and Ablation Analysis in BERT** [↗](#) – *PyTorch, Hugging Face Transformers, scikit-learn, Matplotlib*
 - Analyzed representations across BERT’s 12-layer encoder using linear probing and layer ablation, finding that middle layers provide highly separable representations while upper layers contribute strongly to final predictions in part-of-speech tagging and semantic-similarity tasks.
- **Machine Learning-Based Image Forgery Detection** [↗](#) – *OpenCV, NumPy, SciPy, scikit-learn, Matplotlib*
 - Developed an image-forgery detection pipeline using LBP-DCT features and an ensemble of SVM, Random Forest, and Gradient Boosting classifiers, outperforming the individual classifiers.

COMPETITIONS & ACHIEVEMENTS

Competitive programming & Problems solving

- Solved **1,900+ algorithmic problems** across multiple online judges, participated at **160+ contests**.
- Achieved highest rating of **1611 (Expert)** on Codeforces (@MhD88) [↗](#) and **1730 (3-star)** on CodeChef (@MhD88) [↗](#).
- Top Contest Ranks: Top 2% among 12,275 participants (@MhD88) [↗](#), Top 2% among 16,010 participants (@MhD88) [↗](#)

Hackathons & ML Contests

- Became **Finalist among 400+** teams (Team: JU_NONAME) at Code Samurai Hackathon 2022, hosted by University of Dhaka. Solved algorithmic and SQL problems in preliminaries; developed a web-based system in the final. [↗](#)
- Ranked **1st among 90 teams** (Team: team_hate_mate) at the BLP Workshop 2025 Shared Task [↗](#)

SCHOLARSHIPS & AWARDS

- Received **Excellence Award** twice for contribution in innovation and development at Samsung during Apr-Jun 2025 and Jan-Mar 2024 [↗](#)
- Received an award for achieving the professional-level Software Coding (SWC) certification test on November 08, 2024, in recognition of coding excellence. [↗](#)
- Received **Dean’s Merit scholarships** in all four years of B.Sc. for academic excellence (2018–2023) [↗](#)
- Received prestigious national merit scholarship in both primary and secondary school certification examination (2011, 2013)

TEACHING EXPERIENCES

Competitive Programming Trainer | Department of CSE, Jahangirnagar University 2022 - 2023

- Provided training on C/C++, Data Structures & Algorithms, problem solving strategies
- Organized weekly programming contests; and coordinated annual sessions to attract new students to community.

Academic Tutoring | Private Tutor 2020 - 2024

- Provided academic mentorship to over 80 students across high school, college, and university levels in physics, mathematics, and programming disciplines.

SKILLS

Programming Languages	: C, C++, Java, Kotlin, Python, JavaScript
ML & AI	: scikit-learn, NumPy, OpenCV, PyTorch, Hugging Face Transformers, LLMs
Frameworks & Technologies	: Android, Node.js, Jetpack Compose, HILT, Bootstrap
Databases	: MySQL, MongoDB
Tools	: Git, Perforce, VS Code, Android Studio